0201026/27 2&1 B.B.E.K.O
October 2026
SCIENCE 2&1 **2 &1**
Essay & Objective
2 hours 10 minutes

Name.....

Index Number.....

BEST BRAIN EXAMINATIONS KONSORTIUM
GHANA

Special Mock Examinations for BECE Candidates

October 2026 | **SCIENCE 2 & 1** | 2 hours 10 minutes

ESSAY AND OBJECTIVE

Do **not** *open this booklet until you are told to do so. While you are waiting, read and observe the following instructions carefully. Write your* **name** *and* **index number** *in* **ink** *in the spaces provided above.*

This booklet consists of **two** *papers. Answer Paper* **2** *which comes first, in your answer booklet and Paper* **1** *on your Objective Test answer sheet. Paper* **2** *will last* **1** *hour* **25** *minutes after which the answer booklet will be collected. Do* **not** *start Paper* **1** *until you are told to do so. Paper* **1** *will last* **45** *minutes.*

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PAPER 2 1 hour 25 minutes
ESSAY
[100 marks]

Answer **all** *the questions in your answer booklet*
This paper is in **two** *sections:* **A** *and* **B**. *Answer Question* **1** *in section* **A** *and any other* **three** *questions in section* **B**.
Credit will be given for clarity of expression and orderly presentation of material.

SECTION A
[40 marks]
Answer **all** *the questions in this section.*

1. (a) The electronic structure of an atom is illustrated in Figure 1(*a*).
Study it carefully and answer the questions that follow.

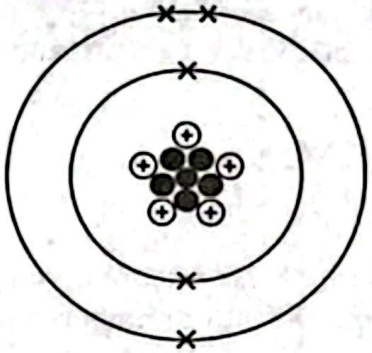


Figure 1*(a)*

(i) State the number of *protons*, *neutrons* and *electrons* in the atom, as well as its electronic configuration.
(ii) Give two differences between electrons and protons.
(iii) Determine the atomic number and mass number of the atom.
(iv) Explain why the atom can be said to be electrically neutral.
(v) What is the name and symbol of the element with the atomic structure above?
[10 marks]

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(b) Figure 1(*b*) is an illustration of two different methods of making a bar magnet.
Study it carefully and answer the questions that follow.

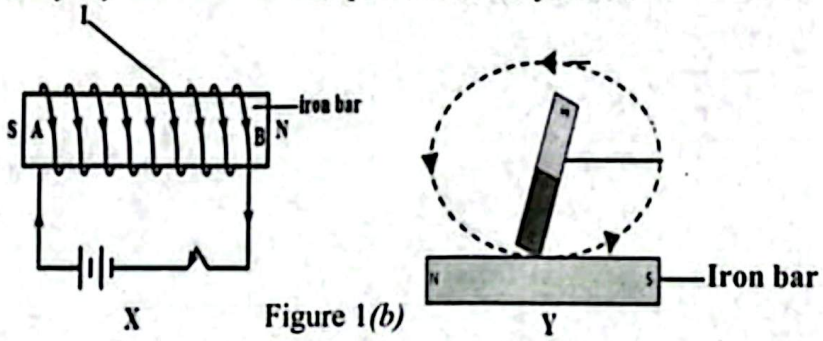


Figure 1*(b)*

(i) Identify the two methods labelled **X** and **Y** above and the type of magnet produced by each.
(ii) State two differences between the two types of magnets produced.
(iii) Outline the steps involved in making a magnet by the method labelled **X**.
(iv) Which two factors determine the strength of the resulting magnet produced at **X**?
(v) Describe how method **Y** is performed.
[10 marks]

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(c) Figure 1(*c*) is an illustration of a food web in an ecosystem.
Study the figure carefully and answer the questions that follow.

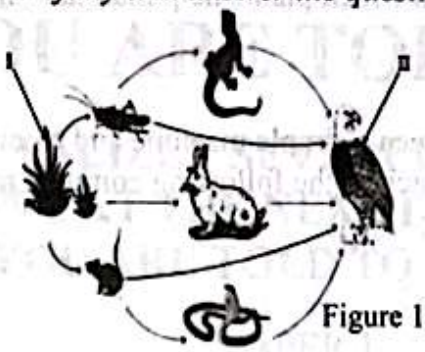


Figure 1*(c)*

(i) Name the primary consumers in the food web above.
(ii) Draw **two** simple food chains from the food web.
(iii) List **two** factors that can interrupt the balance of this ecosystem.
(iv) Describe how the absence of **I** will affect **II**.

[10 marks]

(d) (i) Draw diagrams to illustrate the following seed beds:
(α) Raised seed bed;
(β) Sunken seed bed.

(ii) Rearrange the steps below in the **correct** sequence for preparing a seed bed.

I. Water the soil thoroughly to ensure that it is evenly moist
II. Break up soil, remove weeds, and create furrows for planting.
III. Clear the area of debris, weeds, and other unwanted plant material.
IV. Level the soil, remove rocks and debris, and create a smooth seedbed depth

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(iii) Name one tool used at each of the steps listed in 1(d) (ii) above.
(iv) Give **two** benefits of planting crops on seed beds rather than on flat grounds.

[10 marks]

SECTION B

[60 marks]

Answer **three** *questions* **only** *from this section.*

2. (a) A farmer wants to harvest more than two varieties of crops at a time.
(i) Suggest a farming system he could adopt to get the desired result.
(ii) State **four** reasons why you would suggest this farming system.

[5 marks]

(b) (i) State **one** feature of **each** of the **four** stages in the life cycle of the mosquito.
(ii) State **two** ways to prevent the breeding of mosquitoes.

[6 marks]

(c) (i) Explain how a capacitor differs from a battery.
(ii) State **three** applications of capacitors.

[5 marks]

(d) (i) What is a *suspension* mixture?
(ii) Give **four** examples of suspension mixtures.

[4 marks]

Turn over

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3. (a) (i) Give **two** examples **each** of *biodegradable* and *non-biodegradable* wastes.
(ii) Explain **two** practices used in sustainable waste management.
(iii) State **two** ways in which plants help maintain the balance of carbon dioxide in the atmosphere.

[6 marks]

(b) (i) Distinguish between a simple machine and a complex machine.
(ii) Give the use of each of the following complex machines:
(α) Escalator;
(β) Crane;
(γ) Bicycle.

[5 marks]

(c) Name the **four** types of teeth in mammals and state one function of each. [6 marks]

(d) State the use of each of the following gardening tools:
(i) hand fork;
(ii) trowel;
(iii) knapsack sprayer.

[3 marks]

4. (a) (i) What is *animal feed*?
(ii) List **four** different varieties of animal feed used on the farm.

[5 marks]

(b) (i) Identify **two** structures found in the cytoplasm of a cell.
(ii) State **three** functions of the cytoplasm of a cell.

[5 marks]

(c) State two differences between salt solution and distilled water. [4 marks]

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(d) A boy applies an unbalanced force F to a 120kg box at an acceleration of $0.5m/s^2$
(i) Calculate the force F applied to the box.
(ii) If the box accelerates at $0.3m/s^2$, what is the change in force?

[6 marks]

5. (a) (i) What is *decomposition*?
(ii) Describe the role of decomposition in the carbon cycle.

[4 marks]

(b) (i) Describe how igneous and sedimentary rocks are formed.
(ii) State **two** differences between igneous and sedimentary rocks.

[4 marks]

(c) State the following laws:
(i) Newton's Second Law of Motion;
(ii) The law of magnetism;
(iii) The Law of Conservation of Energy.

[6 marks]

(d) (i) Outline the steps you will take to separate a mixture of sugar and corn flour.
(ii) Name the process involved in each of the steps stated in d (i) above.

[6 marks]

END OF ESSAY TEST

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Answer all the questions.

Each question is followed by **four** *options lettered* **A** *to* **D**. *Find the* **correct** *option for* **each** *question and shade* **in pencil** *on your answer sheet, the answer space which bears the same letter as the option you have chosen. Give only* **one** *answer to* **each** *question. An example is given below.*

Which of the following elements reacts with water?

A. Carbon
B. Sodium
C. Sulphur
D. Iodine

The correct answer is **Sodium**, *which is lettered* **B** *and therefore answer space* **B** *would be shaded.*

[A] [B] [C] [D]

Think carefully before you shade the spaces; erase completely **any** *answer you wish to change.*

Do **all** *rough work on this question paper.*

Now answer the following questions.

1. The type of interatomic bond that involves one atom transferring one or more electrons to another atom is known as
A. covalent bond.
B. ionic bond.
C. molecular bond.
D. atomic bond

2. The water requirement of an animal does **not** depend on its
A. body weight.
B. diet.
C. environmental temperature.
D. physical appearance.

3. Which of the following is **not** true about sedimentary rocks?
A. They occur in layers.
B. They are crystalline in nature.
C. They are relatively soft.
D. They are rounded mineral particles.

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4. The difference between prokaryotic and eukaryotic cells is that
A. Prokaryotic cells have a nucleus, while eukaryotic cells do not.
B. Eukaryotic cells have membrane-bound organelles, while prokaryotic cells do not.
C. Prokaryotic cells are larger than eukaryotic cells.
D. Eukaryotic cells lack DNA, while prokaryotic cells have DNA.

5. The structure of the human teeth helps in chewing food because
I. teeth are hard and durable due to the presence of enamel, which helps to break food down.
II. the different types of teeth are specialized for grinding and tearing food.
III. the shape and arrangement of teeth create an effective biting and chewing mechanism.
A. I, II and III.
B. II and III only.
C. I and III only.
D. I and II only.

6. A fully charged capacitor acts as a/an
A. open circuit.
B. short circuit.
C. resistor.
D. inductor.

7. Which of the following is a homogeneous mixture?
A. Sand and water
B. Oil and water
C. Sea water
D. Muddy water

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8. The part of the human tooth which contains blood vessels is the
   A. pulp cavity.
   B. enamel
   C. dentine.
   D. cement.

9. One property of transition metals is that they have
   A. low density.
   B. good conductivity.
   C. weak molecules
   D. low melting points.

10. The main energy-providing nutrient in animal rations is
   A. proteins.
   B. carbohydrates.
   C. fats.
   D. oils.

11. A common practice in crop rotation to replenish soil nutrients is
   A. planting the same crop continuously for several years.
   B. growing a variety of crops in random order.
   C. leaving the field fallow during the growing season.
   D. alternating the types of crops planted in different years.

12. Plants grow best in a soil that is
   A. acidic
   B. highly acidic
   C. neutral
   D. highly alkaline

13. The correct order of the last four of the first twenty elements is
   A. Chlorine, argon, potassium, calcium.
   B. Chlorine, potassium, argon, calcium
   C. Potassium, chlorine, argon, calcium
   D. Calcium, potassium, chlorine, argon

14. The most effective way to remove plaque from your teeth is by
   A. brushing and flossing.
   B. chewing gum.
   C. using mouthwash.
   D. eating dry fruits and vegetables.

15. In order to complete their life cycle, mosquitoes require a
   A. lot of rainfall.
   B. dirty environment.
   C. stagnant water.
   D. very high temperature.

16. Two liquids with different boiling points can be separated using
   A. distillation
   B. filtration
   C. decantation
   D. centrifugation

17. The physical differences between inner and outer planets relate to
   I. size.
   II. composition.
   III. density.
   A. I and II only
   B. I and III only
   C. II and III only
   D. I, II and III

18. Pigs raised on commercial farms are often fed with a balanced diet that includes
   A. grass and leaves.
   B. fishmeal and algae
   C. corn and soybean meal
   D. wheat and oats

19. Which of the following is **not** typically absorbed by plants from the soil?
   A. Nitrogen
   B. Oxygen
   C. Phosphorus
   D. Potassium

20. In the discharging process of a capacitor, the current flowing through the resistor
   A. increases.
   B. decreases.
   C. remains constant.
   D. stops flowing.

An element has two electrons in the core shell and three electrons in the valence shell of its atom.

Use this information to answer questions **21** *and* **22.**

21. What is the atomic number of the element?
   A. 3
   B. 5
   C. 7
   D. 9

22. The element with this atomic structure is
   A. nitrogen
   B. hydrogen.
   C. oxygen
   D. boron

23. Kerosene and petrol are extracted from crude oil by
   A. distillation.
   B. evaporation.
   C. chromatography.
   D. fractional distillation.

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24. In a second-class lever, the load is located between the
A. fulcrum and wedge.
B. effort and wheel.
C. fulcrum and effort.
D. wheel and axle.

25. Which of the following is a common symptom of communicable diseases?
A. Anaemia
B. Fever
C. Broken bones
D. Muscle cramps

26. Which of the following is a consumer in an ecosystem?
A. Grass
B. Rabbit
C. Sunflower
D. Algae

27. Jacob is feeling severe headache, fever and stiff neck. He is likely suffering from
A. meningitis.
B. Hepatitis.
C. Ebola.
D. COVID.

28. One benefit of planting crops on different seedbeds is that it
A. reduces the need for water.
B. allows farmers to test crop growth in various conditions.
C. eliminates the need for fertilizers.
D. increases the risk of crop failure.

29. Which component in a solar panel traps heat from the solar radiation?
A. Solar thermal turbine
B. Solar thermal collector
C. Solar thermal generator
D. Solar thermal magnet

30. A magnetic force has the ability to
A. cause objects to move.
B. affect stationary objects.
C. speed up motion.
D. cancel out motion.

31. A common prokaryotic organism is
A. humans.
B. bacteria.
C. plants.
D. fungi.

32. Energy conservation helps to
A. reduce the cost of energy.
B. increase the cost of energy.
C. make energy safe to use.
D. encourage the waste of energy.

33. Making fish eat up mosquito larvae as a way to control mosquitoes is a way of
A. chemical control.
B. biological control.
C. genetic control.
D. environmental control.

34. Which of the following objects does **not** possess potential energy?
A. Stretched rubber band
B. A bird perched on a tree branch
C. A mango on a tree
D. An airplane falling from the sky.

35. Which of the following is **not** a pure substance?
A. Water
B. Sugar
C. Air
D. Gold

36. The problem of climate change can be addressed by
I. using more chemical fertilizers.
II. re-using and recycling items.
III. avoiding excessive use of plastics.
IV. using renewable energy sources.
A. I, II, and III only
B. II, III and IV only
C. I, II and IV only
D. I, II, III and IV

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37. Simple carbohydrates in animal feed can be obtained from
A. alfalfa.
B. wheat.
C. bluegrass.
D. orchard grass.

38. The chemical symbol for ozone is
A. H_3O
B. O_4
C. CO_3.
D. O_3

39. The primary goal of a green economy is to
A. ensure economic growth at anyhow.
B. reduce greenhouse gas emissions.
C. increase resource consumption.
D. eliminate poverty.

40. What type of animal feed is composed of fermented corn or grass?
A. Compound feed
B. Blended feed
C. Pellets
D. Silage

END OF PAPER

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